



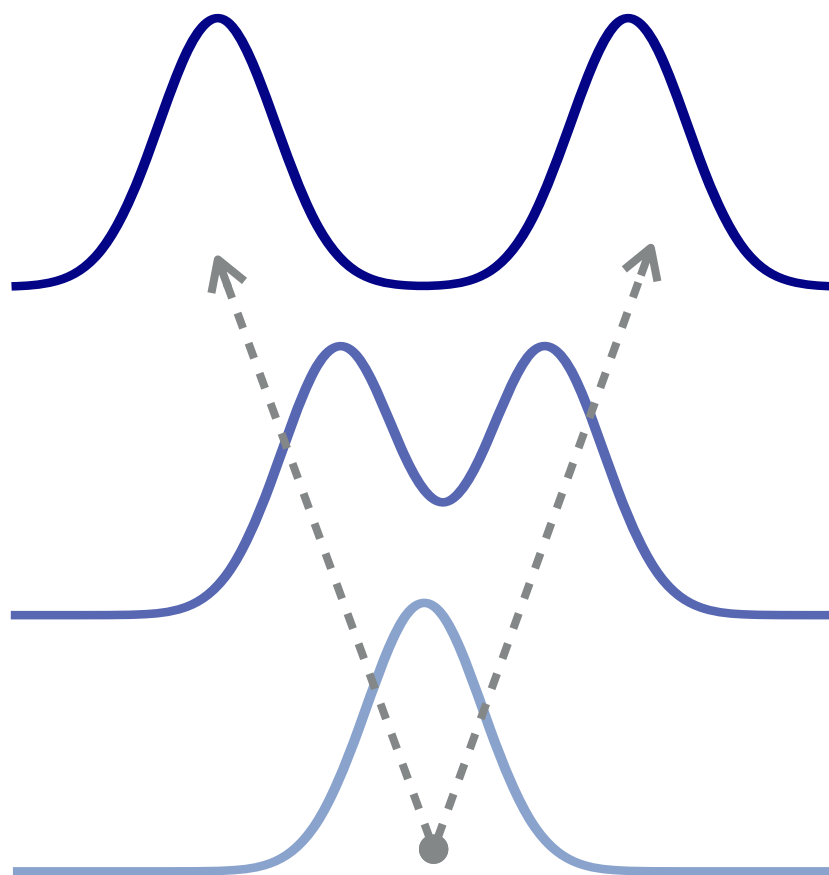


Algorithm: Single chain tries to simultaneously efficiently explore & accurately



Hardware: Clocks are stagnant, and memory computation is distributed

- ▶ **Solution:** Build an interacting particle system
 - ▶ Delegate exploration to tractable **reference** chain
 - ▶ Communicate to **target** through the **interpolating distributions**



Intractable Target

**Interpolating
Distributions**

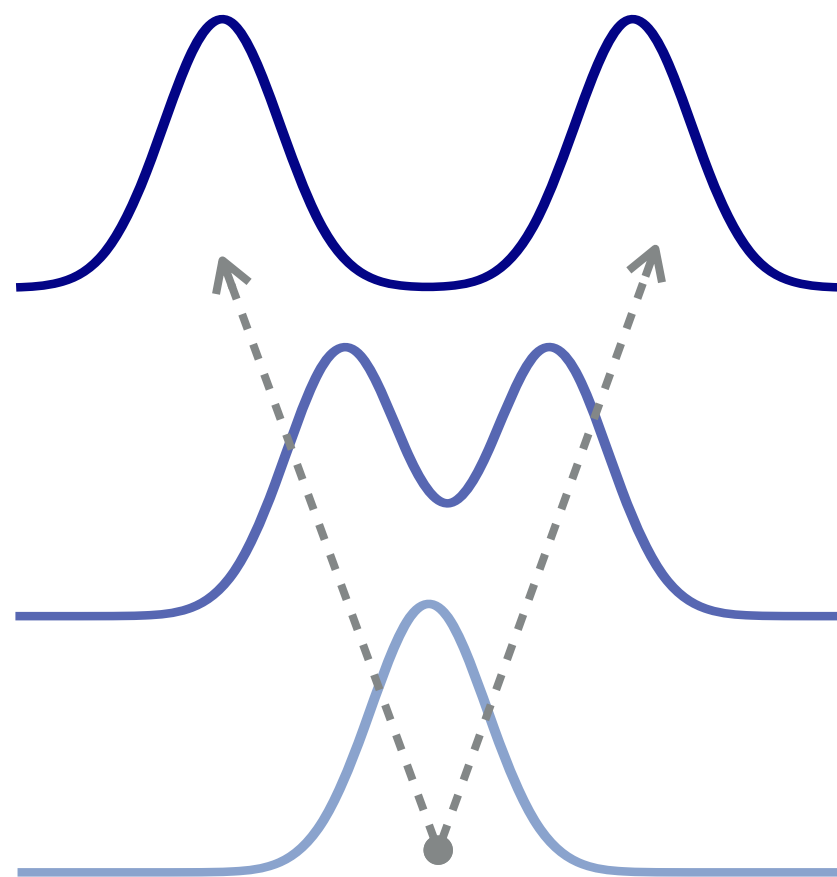
Tractable Reference



PROBLEMS WITH SERIAL MCMC

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- ▶ **Hardware:** Clock speeds are stagnant, and modern computation is distributed
- ▶ **Algorithmic:** Single chain tries to simultaneously efficiently explore & be accurate
- ▶ **Solution:** Build an interacting particle system
 - ▶ Delegate exploration to tractable **reference** chain
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Intractable Target

**Interpolating
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Tractable Reference

ANNEALING ALGORITHMS

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- ▶ Annealing is **not** an algorithm but a platform to build algorithms on top of